Evaluating and Training Large Language Models: What Can (and Cannot) Item Parameters Tell Us?

Research Group Seminar · October 26

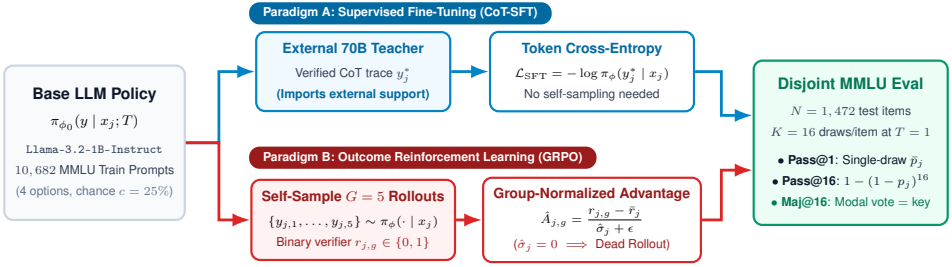
The LLM Evaluation & Post-Training Gap

- **Evaluation Gap**: Outcome RL (GRPO) beats CoT-SFT on single-draw $\mathrm{Pass@1}$ ($\mathbf{47.9}\%$ vs. $\mathbf{43.9}\%$), yet collapses on $\mathrm{Pass@16}$ ($\mathbf{65.6}\%$ vs. $\mathbf{96.6}\%$) and gains $\mathbf{0}\%$ on Majority Vote ($\mathrm{Maj@16} = \mathbf{47.5}\%$).
- **Naive 2PL Readout Fails**: Plugging 102-LLM 2PL (a_j, b_j) into adaptive testing (CAT) hits a $\mathbf{2.4}\%$ bias floor and adds $\mathbf{0}\%$ gain in RL critics.

What Item-Level Rollout Diagnostics Reveal

- **1. Standing $\neq$ Plasticity**: Cross-model a_j skews $5\times$ toward memorization and is orthogonal to RL updates ($\mathbf{r = 0.072}$).
- **2. Consolidation vs. Support Shift**: RL sharpens slope ($\tau = 1.85, l_z = +\mathbf{7.55}$); SFT shifts support but causes easy-item inversions ($l_z = -\mathbf{16.99}$).
- **3. Guessing & Distractors**: $\mathbf{53.9}\%$ of active RL prompts sit at $I_j^{\mathrm{3PL}} = 0$ ($\leq 25\%$); wrong modal distractors dominate $\mathbf{51.4}\%$ of items.

Background: LLM Post-Training & Multi-Draw Evaluation

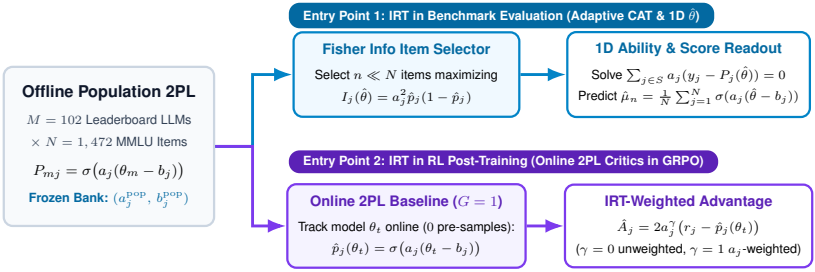


CoT-SFT (Teacher Trace)
Minimizes $-\log \pi_\phi(y_j^* \mid x_j)$ on verified 70B teacher traces.
→ **Overwrites wrong modal beliefs** even when Base pass rate $p_j = 0$.

GRPO (Binary Outcome RL)
Rewards $\{0, 1\}$ correctness across $G = 5$ self-rollouts ($\hat{\sigma}_j = \sqrt{p_j(1 - p_j)}$).
→ **Blind to lucky guesses** at $\leq 25\%$ and **which wrong distractor** is modal.

Rollout Readouts ($K = 16$ draws)
- $\mathbf{Pass@1}$: Mean single-draw accuracy.
- $\mathbf{Pass@16}$: Coverage across 16 draws.
- $\mathbf{Maj@16}$: Whether the *modal option* is the correct key ($p_{\mathrm{key}} = p_{\mathrm{top}}$).

Background: How Static 1D 2PL Enters Evaluation & RL



Entry 1: Adaptive Evaluation (CAT)

Calibrate 1D binary 2PL (a_j, b_j) across $M = 102$ leaderboard LLMs (Maia Polo et al., 2024; Kipnis et al., 2025).

$\rightarrow$ Select items maximizing $I_j(\hat{\theta}) = a_j^2 \hat{p}_j(1 - \hat{p}_j)$ to estimate model ability from $n \ll N$ items.

Entry 2: Online 2PL Critics in RL

Replace $G = 5$ group rollouts with a single rollout ($G = 1$) using static 2PL expectation $\hat{p}_j(\theta_t) = \sigma(a_j(\theta_t - b_j))$:

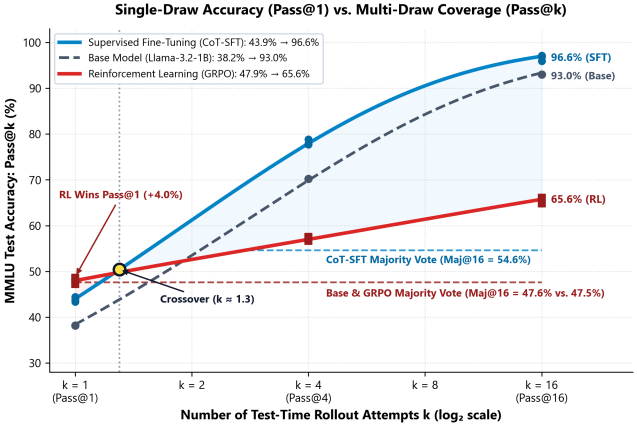
$$\hat{A}_j = 2\,a_j^{\gamma}\big(r_j - \hat{p}_j(\theta_t)\big), \qquad \gamma \in \{0, 1\}$$

Terminology Bridge: LLM Rollouts & Psychometric Constructs

Concept	LLM Rollout Diagnostic	Psychometric / IRT Construct	Key Metric
1. Item Baseline	Base initial pass rate ($p_{\text{base},j}$)	Model-specific item log-odds (μ_{0j})	$\mu_{0j} = \text{logit}(p_{\text{base},j})$
2. Sharpening	Inverse temperature ($1/T$) or RL	Within-model dispersion slope (τ)	$\tau \leftrightarrow 1/T$
3. Consistency	Easy-item misses vs. step curve	Drasgow Person-Fit & Guttman order	$l_z = +7.55$ vs. -16.99
4. Guessing Trap	Active GRPO variance at $\leq 25\%$	2PL Variance vs. 3PL Information	$I_j^{\text{3PL}} = 0\ (\leq 25\%)$
5. Distractor Mode	Modal option ($p_{\text{top},j}$) vs. key ($p_{\text{key},j}$)	Nominal Response Model (Bock, 1972)	$p_{\text{top},j} > p_{\text{key},j}$ (51.4%)
6. Prompt Bias	Option-order gap (`A/B/C/D` swap)	Differential Item Functioning (DIF)	$D = \lvert p_{\text{ord0}} - p_{\text{ord1}} \rvert$

Diagnostic Strategy: Without any retraining, item-level rollout statistics ($p_{\text{base},j}$, $p_{\text{top},j}$, l_z, I_j^{3PL}) expose *why* single-draw evaluation and static 1D 2PL fail—and why SFT and RL play complementary roles.

The Evaluation Paradox: Single-Draw Pass@1 vs. Multi-Draw Coverage



1. Single-Draw Pass@1 Illusion ($k = 1$)

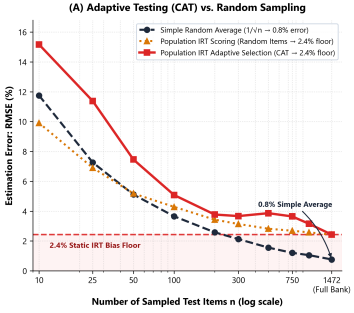
- Base: $\mathbf{38.2\%}$ | CoT-SFT: $\mathbf{43.9\%}$
- GRPO (RL): $\mathbf{47.9\%}$ (appears $+4.0\%$ ahead of SFT at $k = 1$).

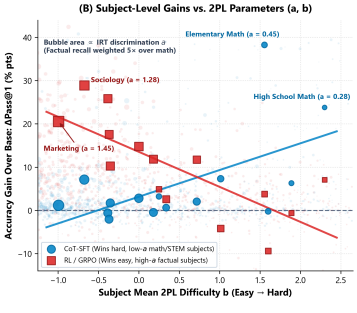
2. Reversal on Pass@16 & Maj@16

- **Coverage collapse**: GRPO drops to $\mathbf{65.6\%}$ vs. Base $\mathbf{93.0\%}$ and CoT-SFT $\mathbf{96.6\%}$.
- **Zero modal gain under RL**: GRPO Maj@16 ($\mathbf{47.5\%}$) = Base ($\mathbf{47.6\%}$), whereas CoT-SFT reaches $\mathbf{54.6\%}$.

→ GRPO only sharpens existing Base modes; CoT-SFT flips modal answers to the correct key.

Evaluation Gap: Adaptive Testing & Memory vs. Reasoning Skew





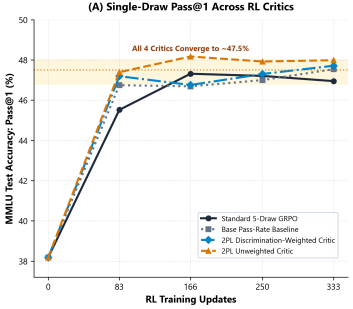
(A) Adaptive CAT Hits a 2.4% Bias Floor

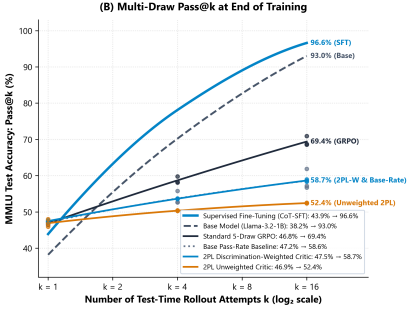
- Simple random sampling reaches $\mathbf{0.8}\%$ RMSE at full bank.
- 1D IRT scoring & CAT plateau at $\mathbf{2.4}\%$ error because post-trained models violate 1D item invariance.

(B) $5\times$ Discrimination Skew Across Subjects

- Factual recall has $5\times$ higher a_j (*Marketing* $\mathbf{1.45}$) than math (*HS Math* $\mathbf{0.28}$). Since $I_j \propto a_j^2$, CAT over-samples memory $25\times$.
- GRPO gains on easy/high-a items; CoT-SFT gains on hard/low-a math.

Training Gap: Why Static 2PL RL Critics Saturate





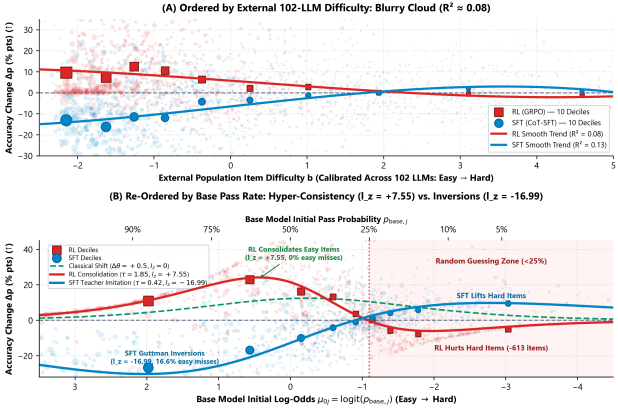
(A) All 4 Critics Saturate at $\sim 47.5\%$ Pass@1

- Standard `GRPO` ($G = 5$) and single-rollout ($G = 1$) 2PL critics all converge to $\mathbf{46.8\%}$–$\mathbf{47.5\%}$.
- Online 2PL baseline cuts sampling cost $5\times$, but cross-model a_j-weighting adds **zero accuracy gain**.

(B) High-k Collapse & Item-Subset Divergence

- At $k = 16$, all RL critics collapse ($\mathbf{52.4\%}$–$\mathbf{69.4\%}$) far below `Base` ($\mathbf{93.0\%}$).
- Despite tied aggregate Pass@1, **5 of 6 critic pairs** learn statistically distinct item subsets ($p < 0.05$).

Coordinate Shift: Cross-Model Standing vs. Base Pass Rate



1. Why Static 2PL Critics Fail (Panel A)

Ordering item gains Δp_j by 102-LLM difficulty b_j gives a flat cloud ($\mathbf{R^2 = 0.08}$).

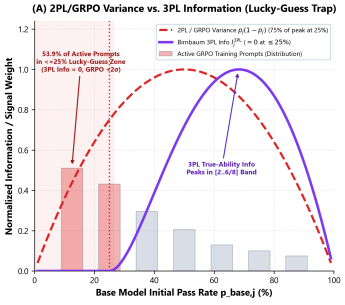
$\rightarrow$ Gradient updates are orthogonal to cross-model a_j ($\mathbf{r = 0.072}$): **measuring standing $\neq$ receiving change**.

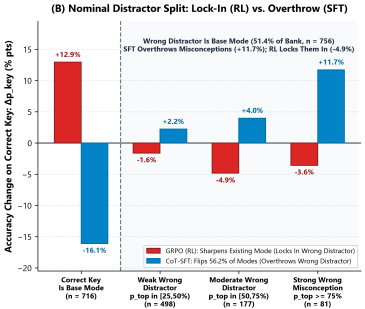
2. Base Frame (μ_{0j}) & Person-Fit l_z (Panel B)

Re-ordering by Base log-odds $\mu_{0j} = \text{logit}(p_{\text{base},j})$ ($\mathbf{R^2 = 0.35}$) reveals opposite **Drasgow Person-Fit (l_z)** regimes:

- **RL Consolidation** ($\tau = 1.85, l_z = +7.55$): Hyper-consistent step function (0% easy misses), but hurts 613 hard items ($< 25\%$).
- **SFT Imitation** ($\tau = 0.42, l_z = -16.99$): Lifts hard items, but causes 16.6% easy-item Guttman inversions.

Training Gap: Lucky-Guess Trap & Attractive Distractors





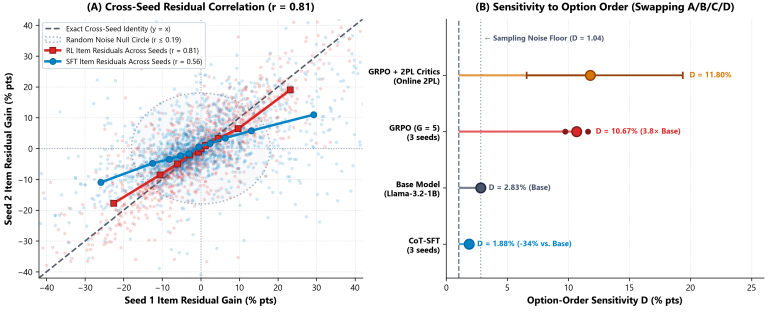
(A) 3PL Lucky-Guess Trap ($I_j^{3\mathrm{PL}} = 0$ at $\leq 25\%$)

- At $c = 25\%$, Birnbaum's 3PL information is $\mathbf{0}$, yet 2PL/GRPO variance $p_j(1-p_j)$ is still $\mathbf{75\%}$ of peak.
- $\mathbf{53.9\%}$ of active GRPO prompts $(4,468/8,296)$ sit in $\leq 25\%$, wasting updates on lucky guesses $(+2\sigma)$.

(B) Nominal Distractor Split ($p_{\mathrm{top},j} > p_{\mathrm{key},j}$)

- On $\mathbf{51.4\%}$ of the bank $(756$ items), a wrong distractor is Base's mode ($\bar{p}_{\mathrm{top}} = \mathbf{48.0\%} > \bar{p}_{\mathrm{key}} = \mathbf{19.6\%}$).
- **RL locks in wrong modes** (-4.9%); **CoT-SFT flips 56.2% of modes** $(+11.7\%$ on strong misconceptions).

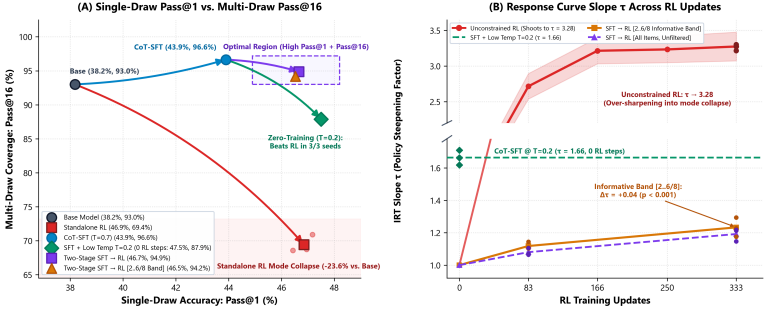
Downstream Symptoms: Cross-Seed Residuals & Option-Order Bias



(A) Cross-Seed Residual Correlation ($\mathbf{r = 0.81}$)
After removing the 1D curve, item residuals across random seeds remain highly correlated ($\mathbf{r = 0.81}$ in RL vs. 0.19 noise floor).
$\rightarrow$ Item-level learning is deterministic and multi-dimensional, locked to Base initialization.

(B) RL Amplifies Option-Order Bias $4\times$
Swapping option letters (`A/B/C/D`) measures position sensitivity D:
- `Base`: $\mathbf{2.83\%}$ | `CoT-SFT`: $\mathbf{1.88\%}$ (-34% bias)
- `GRPO`: $\mathbf{10.67\%}$ ($4\times$ **position bias** when sharpening low-confidence items).

Immediate Interventions: Temperature, Two-Stage SFT→RL & 3PL Band



1. Temperature Shortcut
Since RL acts like $\tau \leftrightarrow 1/T$, decoding CoT-SFT at $\mathbf{T = 0.2}$ (**0 RL steps**) jumps to $\mathbf{47.5}\%$ Pass@1 while keeping $\mathbf{87.9}\%$ Pass@16.

2. Two-Stage SFT → RL
SFT first overthrows wrong distractors ($p_{\text{top}} > p_{\text{key}}$); RL then repairs SFT's easy-item inversions ($l_z = -16.99 \rightarrow \mathbf{94.9}\%$ Pass@16).

3. 3PL Band $[2..6/8]$
Dropping the $1/8$ lucky-guess bin ($I_j^{\text{3PL}} = 0$) cuts dead rollouts $\mathbf{34}\% \rightarrow \mathbf{23}\%$ and steepens τ ($+\mathbf{0.04}, \mathbf{p < 0.001}$).

Summary: LLM Evaluation/Training Gaps & What Our Diagnostics Bring

1. Existing Gaps in LLM Eval & Training

- **Single-Draw Evaluation Illusion**: $\mathrm{Pass@1}$ confounds policy sharpening ($\tau \leftrightarrow 1/T$) with knowledge growth ($\Delta\theta$), masking RL's $\mathbf{0}\%$ modal gain ($\mathrm{Maj@16} = 47.5\%$) and $-\mathbf{23.6}\%$ coverage collapse.
- **Binary RL Training Blindness**: Outcome RL (`GRPO`) wastes $\mathbf{53.9}\%$ of active updates in the $\leq 25\%$ lucky-guess zone ($I_j^{\mathrm{3PL}} = 0$) and locks in wrong modal distractors ($p_{\mathrm{top}} > p_{\mathrm{key}}$ on $\mathbf{51.4}\%$ of items, $-\mathbf{4.9}\%$).

2. Why Naive 1D 2PL Readouts Cannot Fix Them

- **Domain Skew in Evaluation**: Cross-model 102-LLM a_j weights memorization $5\times$ higher than math (1.45 vs. 0.28), causing a $\mathbf{2.4}\%$ `CAT` error floor.
- **Standing $\neq$ Plasticity in RL**: Cross-model a_j is orthogonal to within-model gradient updates ($\mathbf{r} = \mathbf{0.072}$); all 4 RL critics saturate at $\sim \mathbf{47.5}\%$.
- **Binary $\{0, 1\}$ Misspecification**: Standard 2PL ignores guessing ($c = 25\%$), attractive distractors, and policy temperature τ.

3. What Our Rollout Diagnostics Bring

- **Zero-Retraining Decomposition**: Separates **RL Guttman Consolidation** ($\tau = 1.85, l_z = +\mathbf{7.55}$, 0% easy misses) from **SFT Distractor Overthrow** ($\mathbf{56.2}\%$ mode flips, $+\mathbf{11.7}\%$ on strong misconceptions, $l_z = -\mathbf{16.99}$).
- **Actionable Gains**: Enables `CoT-SFT @ T=0.2` ($\mathbf{47.5}\%$ $\mathrm{Pass@1}$ in 0 RL steps), `Two-Stage SFT→RL` ($\mathbf{94.9}\%$ $\mathrm{Pass@16}$), and `3PL Band` $[2..6/8]$ ($\mathbf{34}\% \rightarrow \mathbf{23}\%$ dead rollouts).

Next Steps I: Why Static Rollouts & Naive Dynamic Models Fail

1. Non-Ergodicity & Chain-Rule Dynamic Discrimination

Static rollout diagnostics ($\hat{p}_{j,0}, l_z, I_j^{\mathrm{3PL}}$) are single-checkpoint snapshots. Along policy trajectory $\boldsymbol{\phi}_t$ with velocity $\boldsymbol{u}_t = \dot{\boldsymbol{\phi}}_t$:

$$a_j^{\mathrm{dyn}}(t) = \frac{\partial \mathrm{logit}\, P_j}{\partial t} = \underbrace{\nabla_{\boldsymbol{\phi}} \mathrm{logit}\, P_j^{\top}}_{\text{Jacobian } \boldsymbol{J}_j} \underbrace{\boldsymbol{u}_t}_{\text{Velocity}} \perp a_j^{\mathrm{pop}}$$

Orthogonal to static a_j^{pop} ($r = 0.072$); logit evolution splits into **slope** τ_t (Ferrando, 2007), **shift** δ_t, and **item drift** $\kappa_{t,j} = -\Delta b_{t,j}$ (Haebara, 1980).

2. The $\mathcal{O}(\Delta^2)$ Second-Order Sink Principle

By score expansion around the unperturbed trajectory:

$$\hat{\theta}_W - \theta_W^* \approx \mathcal{J}_{W\cdot o}^{-1} \sum_{t=1}^{T} \mathbb{E}\big[p_t(1-p_t)\,\delta_t\, g_t\big]$$

Fine-grained transitions enter 1-bit likelihoods at **second order** $\mathcal{O}(\Delta^2)$, acting as a **sink** for unclosed $\mathcal{O}(1)$ nuisances (format, τ_t, baseline standing). Identification requires **closing all** $\mathcal{O}(1)$ **channels first** and training with **masked multi-step open-loop horizons** ($H > 1$).

3. Subspace Absorption, Dual-Role & Routing Laws

$$\mathcal{P}_{\perp}(\boldsymbol{W}) = (\boldsymbol{I} - \boldsymbol{A}\boldsymbol{A}^{+})\boldsymbol{W}, \qquad \boldsymbol{a}_j^{\mathrm{read}} \in \Delta^{M-1} \neq \boldsymbol{\psi}_j^{\mathrm{write}}$$

- **In-Span Absorption**: Unconstrained global drift $\boldsymbol{v}_t$ absorbs $\mathrm{col}(\boldsymbol{A})$; only $(\boldsymbol{I} - \boldsymbol{A}\boldsymbol{A}^{+})\boldsymbol{W}$ is identifiable.
- **Dual-Role Collapse**: Sharing one simplex vector $\boldsymbol{a}_j$ for static covariance and dynamic updates collapses skill clusters into single-item memorization ($\max_m a_{j,m} : 0.38 \to 0.64$).
- **Closed-Loop Routing (Forssell & Ljung, 1999)**: Omitting any adaptive routing dimension manufactures spurious transfer ($r = 0.607$).

4. Why White-Box LLMs Solve Passive-Log Limits

Passive 1-bit human logs (8–11 events/student) hit van Trees minimax bounds. **LLMs literally are gradient learners** with readable option distributions $\boldsymbol{p}_{j,t} \in \Delta^3$ and forkable checkpoints for exact $\mathrm{do}(\text{train on } A)$.

Next Steps II: Mathematical Derivation of the Dynamic State-Space Model

1. Hierarchical Places & Parallel Prefix-Sum ADF

$$P_{j,t} = \sigma\Big(\tau_t\big(\underbrace{\boldsymbol{m}_j^\top \boldsymbol{v}_t}_{\text{Global } K} + \underbrace{l_j \boldsymbol{a}_j^\top \boldsymbol{z}_t}_{\text{Places } M} + \underbrace{\boldsymbol{u}_j^\top \boldsymbol{c}_t}_{\text{Plasticity}} - b_j\big) + r_\psi\Big)$$

- Learn frozen simplex **Read Keys** $\boldsymbol{a}_j \in \Delta^{M-1}$ ($M = 256$) from $\mathcal{O}(1)$ static residual covariance ($r = 0.81$) via **Information-Form ADF** in $\mathcal{O}(MK + K^3)$ using **GPU parallel prefix-sum associative scans**.
- **Zero-Init Residual** $r_\psi(\boldsymbol{h}_t, \boldsymbol{e}_j)|_{t=0} \equiv 0$ + **Stochastic Scale-Orbit Gate** $\tilde{z}_k = \sigma((\rho_k + \nu_k)/T_z)$ prunes redundant dimensions.

2. Doob Martingale Split & Two Structural Zeros

$$\Delta\boldsymbol{\kappa}_t = \underbrace{(\boldsymbol{\kappa}_{t+1} - \mathbb{E}[\boldsymbol{\kappa}_{t+1} \mid \mathcal{F}_t])}_{\text{Epistemic } \Delta\boldsymbol{z}_t:\ \mathbb{E}[\Delta\boldsymbol{z}_t]=\mathbf{0}} + \underbrace{(\mathbb{E}[\boldsymbol{\kappa}_{t+1} \mid \mathcal{F}_t] - \boldsymbol{\kappa}_t)}_{\text{Answer-Blind Drift } \Delta\boldsymbol{c}_t \text{ (No } y_t)}$$

- **Zero 1 (Epistemic)**: 3PL-aware filter updates *only* standing $(\boldsymbol{v}_t, \boldsymbol{z}_t)$ and shrinks $\boldsymbol{\Sigma}_t$ from residual $(y_t - \bar{p}_t)$.
- **Zero 2 (Ontological)**: Transition $\Delta\boldsymbol{c}_t$ is **answer-blind** (receives $\bar{p}_t, \boldsymbol{\psi}_{e_t}$, never y_t), blocking $\mathcal{O}(1)$ residual leakage.

3. Interior Step Law, Preconditioner & Convex Gate

Policy-gradient covariance $\mathrm{Cov}_{\pi_\phi}(r, \nabla_\phi \log \pi_\phi) = p_j(1 - p_j)\boldsymbol{\psi}_j$ maps to the **3PL Interior Plasticity Law**:

$$u(\bar{p}_t) = c\,(\bar{p}_t - c_0)_+^{k_1}(1 - \bar{p}_t)^{k_2}, \qquad \boldsymbol{s}_{t+1} = \boldsymbol{s}_t + \boldsymbol{g}_t^2$$

$$\boldsymbol{c}_{t+1} = (\boldsymbol{I} - \boldsymbol{\Lambda})\,\boldsymbol{c}_t + u(\bar{p}_t)\,(1 + \boldsymbol{s}_t)^{-\beta} \odot \left(\boldsymbol{L}\boldsymbol{L}^\top \boldsymbol{\psi}_{e_t}\right)$$

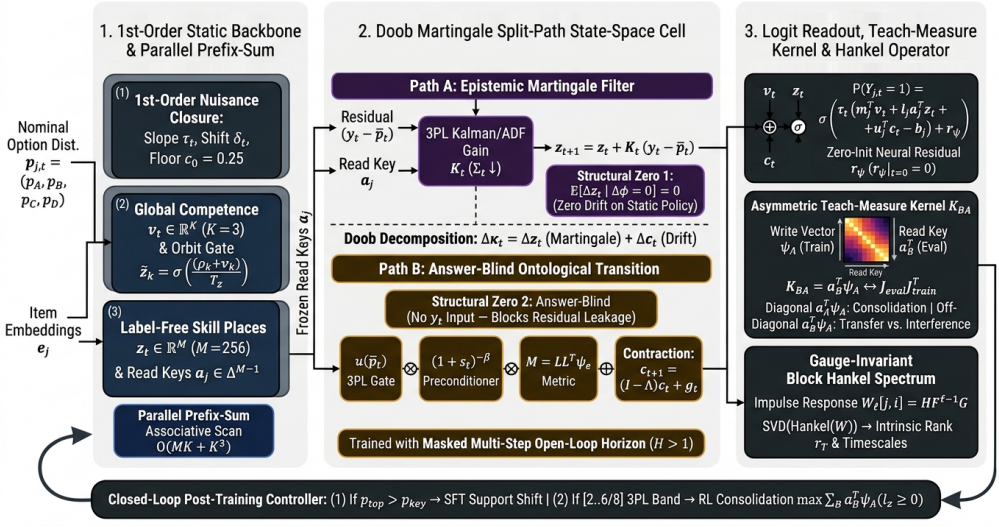
- Unifies SGD ($k = (1, 1)$), std-norm RL ($k = (\frac{1}{2}, \frac{1}{2})$), and 3PL floor $c_0 = 0.25$ ($[2..6/8]$ band); $(1 + \boldsymbol{s}_t)^{-\beta}$ tracks Adam saturation; convex gate $\boldsymbol{\Lambda} = \mathrm{diag}(\lambda_m) \in (0, 1)^M$ prevents Jacobian explosion.

4. Asymmetric Teach–Measure Kernel & Hankel SVD

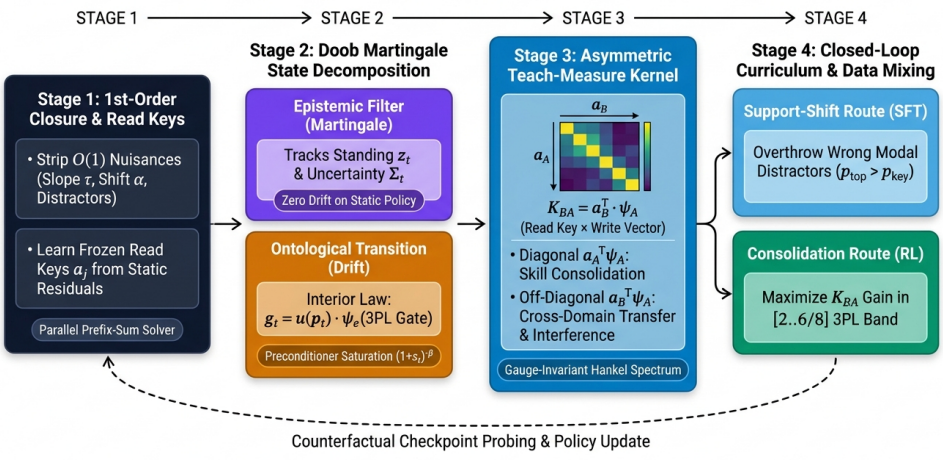
$$K_{BA} = \boldsymbol{a}_B^\top \boldsymbol{\psi}_A \leftrightarrow \boldsymbol{J}_{\text{eval}} \boldsymbol{J}_{\text{train}}^\top, \quad W_\ell[j, i] = \boldsymbol{H}\boldsymbol{F}^{\ell-1}\boldsymbol{G}$$

- Unties **Read Key** $\boldsymbol{a}_B$ from **Write Vector** $\boldsymbol{\psi}_A$: diagonal $\boldsymbol{a}_A^\top \boldsymbol{\psi}_A$ measures consolidation; off-diagonal $\boldsymbol{a}_B^\top \boldsymbol{\psi}_A$ predicts transfer vs. interference.
- $\mathrm{SVD}(\mathrm{Hankel}(W_1, \ldots, W_m))$ extracts gauge-invariant update rank r_T and memory timescales under $GL(K)$ rotations.

Next Steps III: Internal Architecture of the Dynamic State-Space Cell



Next Steps IV: Closed-Loop Curriculum & Data-Mixing Pipeline



Thank You!

Backup B1: First-Order Expansion & Full Results

First-Order Logit Expansion

For policy log-odds $z_j(\phi) = \operatorname{logit} p_j(\phi)$ under parameter update $\Delta\phi = \alpha_\parallel u_{\text{pop}} + u_\perp$:

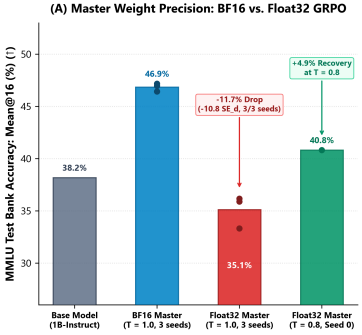
$$\Delta p_j \approx p_{0j}(1 - p_{0j})\left[\alpha_\parallel a_j^{\text{pop}} + J_j u_\perp\right]$$

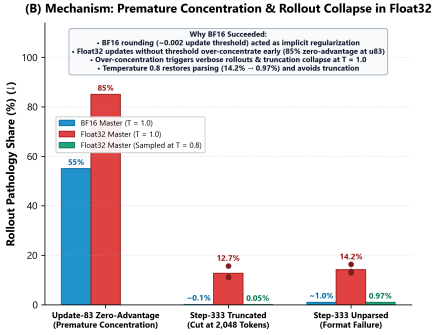
- $\operatorname{corr}(J_j u_\perp, a_j^{\text{pop}}) = \mathbf{0.072}$.
- Within-model variance $p_{0j}(1 - p_{0j})$ drives Δp_j ($R^2 = 0.354$), while a_j^{pop} has near-zero alignment ($r = +0.025$).

Disjoint MMLU Test Bank ($N = 1,472$)

Model / Arm	Draws	Pass@1	Pass@16	Maj@16	Order D
Base (Llama-3.2-1B)	—	38.2%	93.0%	47.6%	2.83%
CoT-SFT (70B Teacher)	0	43.9%	**96.6%**	**54.6%**	**1.88%**
GRPO Standard ($G = 5$)	5	46.9%	69.4%	47.5%	10.67%
Base Pass-Rate Baseline	1	47.2%	58.6%	47.4%	11.42%
2PL Unweighted ($\gamma = 0$)	1	46.9%	52.4%	47.1%	12.15%
2PL a_j-Weighted ($\gamma = 1$)	1	**47.5%**	58.7%	47.5%	11.80%
CoT-SFT @ T=0.2 (0 RL)	0	**47.5%**	87.9%	54.6%	1.95%
Two-Stage SFT → RL	1	46.7%	94.9%	54.4%	3.12%
Two-Stage [2..6/8 Band]	1	46.5%	94.2%	54.5%	2.98%

Backup B2: Numerical Precision Audit (BF16 vs. Float32)





(A) Float32 Master Weights Drop Accuracy by -11.7%

Removing BF16 rounding (~ 0.002 threshold) lowers Pass@1 from $\mathbf{46.9}\% \rightarrow \mathbf{35.1}\%$ (partially recovered at $T = 0.8$).

(B) Premature Concentration in Float32

Exact Float32 updates over-concentrate early ($\mathbf{85}\%$ zero-variance rollouts vs. 55% in BF16), causing truncation (12.7%) and parse errors (14.2%).